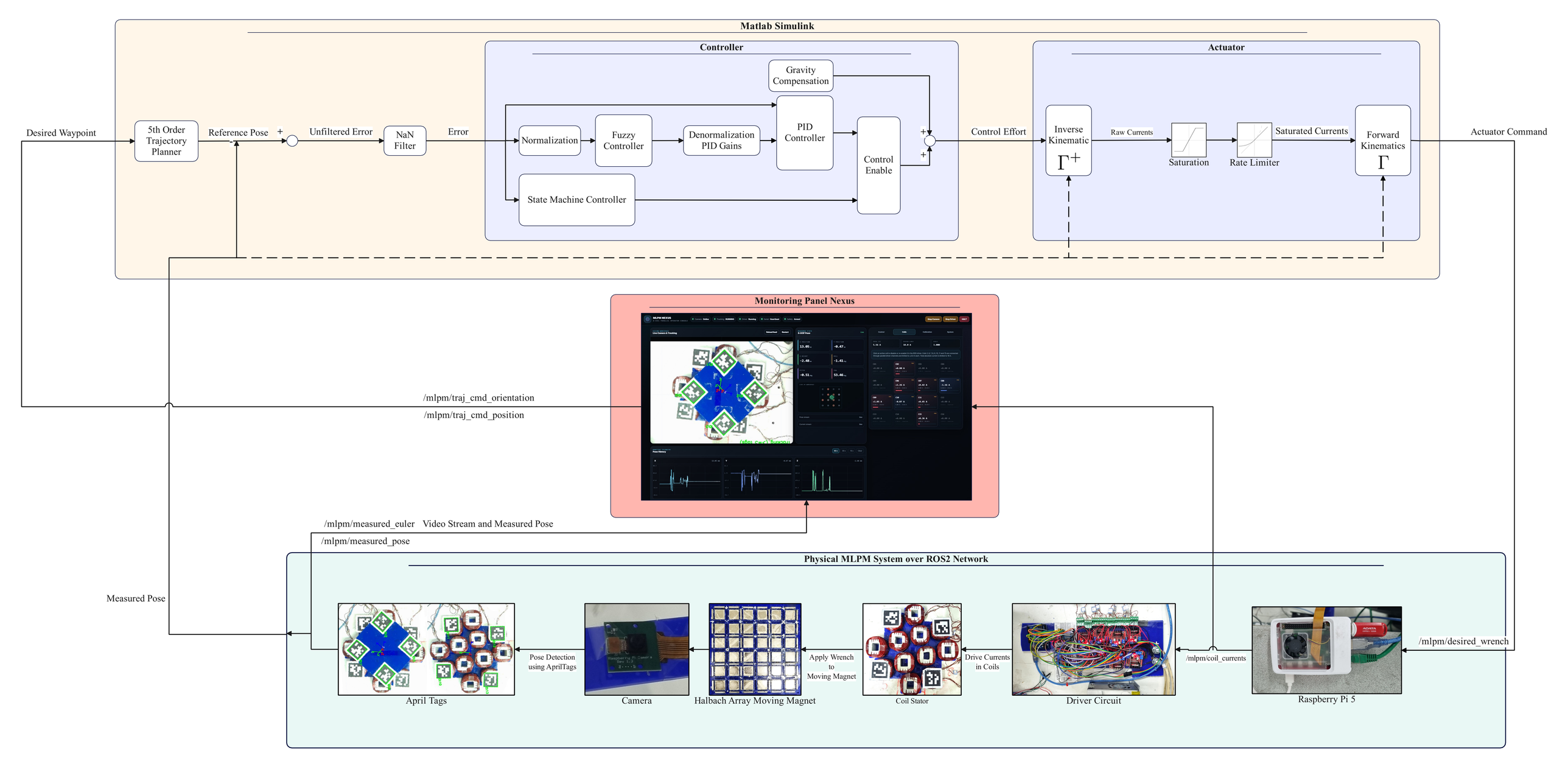

Matlab Simulink
Controller
Actuator
Desired Waypoint
5th Order Trajectory Planner
Reference Pose
+
-
Unfiltered Error
NaN Filter
Error
Gravity Compensation
Normalization
Fuzzy Controller
Denormalization PID Gains
PID Controller
Control Enable
State Machine Controller
+
+
Control Effort
Inverse Kinematic
Γ+
Raw Currents
Saturation
Rate Limiter
Saturated Currents
Forward Kinematics
Γ
Actuator Command
Monitoring Panel Nexus
/mlpm/traj_cmd_orientation
/mlpm/traj_cmd_position
/mlpm/measured_euler
Video Stream and Measured Pose
/mlpm/measured_pose
Measured Pose
Physical MLPM System over ROS2 Network
April Tags
Pose Detection using AprilTags
Camera
Halbach Array Moving Magnet
Apply Wrench to Moving Magnet
Coil Stator
Drive Currents in Coils
Driver Circuit
/mlpm/coil_currents
Raspberry Pi 5
/mlpm/desired_wrench